Hugo Marquerie 22/01/2026

Lem involucion disco unidad rho

Lema 1. *Sea* $f = \tau_a \circ \rho_\gamma \in \operatorname{Aut}(\mathbb{D})$, *entonces* $f = \rho_\gamma \circ \tau_{\overline{\gamma} a}$.

***Demostración*:** Como $\gamma \in \mathbb{T}$, tenemos que $\gamma \cdot \overline{\gamma} = |\gamma|^2 = 1$. Por tanto,

$$\forall z \in \mathbb{D} : f(z) = \tau_a(\rho_\gamma(z)) = \frac{a - \gamma z}{1 - \overline{a}\gamma z} \cdot \frac{\overline{\gamma}}{\overline{\gamma}} = \frac{1}{\overline{\gamma}} \cdot \frac{\overline{\gamma} a - z}{1 - \overline{a}\gamma z} = \gamma \cdot \frac{\overline{\gamma} a - z}{1 - \overline{(\overline{\gamma} a)} z} = \rho_\gamma(\tau_{\overline{\gamma} a}(z)).$$

■

Referenciado en

- `ejer-aut-disco-unidad-inversa`
- `teo-aut-disco-unidad-composicion`

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